4.1. Asymptotic distribution & statistical behavior of orbits.

$f : X \to X$: continuous

$$I_x : C(X) \to \mathbb{R}$$ time average

$$I_x(\varphi) := \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi(f^k(x))$$

- linearity : $I_x(\alpha\varphi + \beta\psi) = \alpha I_x(\varphi) + \beta I_x(\psi)$
- bdd : $|I_x(\varphi)| \le \sup|\varphi(y)|$
- positivity : $I_x\varphi \geq 0$ if $\varphi \geq 0$, $I_x 1 = 1$.
- inv. under f : $I_x(\varphi\circ f) = I_x(\varphi)$ $(= I_{f(x)}(\varphi))$.

$\hookrightarrow \mu(f^{-1}A) = \mu(A)$

Riesz rep. Thm,

pos. bdd. linear functional $\Rightarrow \exists!$ Borel probability measure μ.

i.e. $I_x(\varphi) = \int_X \varphi \, d\mu$.

$\exists!$ BP measure μ.

Main questions :

- $\exists?\ x \in X$ s.t. asymptotic distributions I_x exists.
- f-invariant measure $\mu \Rightarrow$ Can we "Find" such $x \in X$?

  i.e. $\exists?\ x\in X$ s.t. $\int \varphi \, d\mu = I_x(\varphi)$.

Thm. (Krylov - Bogolubov).

Any contr. on a cpt metrizable sp. X has an inv. BP measure.

sketch) $\mu_n := \frac{1}{n}\sum_{k=0}^{n-1} \delta_{f^k(x)}$: probability measure.

$\Rightarrow \int \varphi \, d\mu_n = \frac{1}{n}\sum \varphi(f^k(x))$

$\exists$ convergent subseq. $\mu_{n_j} \to \mu$ (Banach-Alaoglu Thm).

$\varphi \in C(X) \Rightarrow \int \varphi\circ f \, d\mu_n - \int \varphi \, d\mu_n = \frac{\varphi(f^n(x)) - \varphi(x)}{n} \to 0$

$\Rightarrow \int \varphi\circ f \, d\mu = \int \varphi \, d\mu$: f-invariant measure.

weak* conv in $C(X)^*$ $\mu_n \to \mu \iff \int \varphi \, d\mu_n \to \int \varphi \, d\mu \quad \forall \varphi \in C(X)$.

Another) $x \in X$, $\{\varphi_1, \varphi_2, \cdots\}$: Countable dense in $C(X)$.

For each m, $\frac{1}{n}\sum_{k=0}^{n-1} \varphi_m(f^k(x))$: bdd.

$\exists$ conv. subseq $\Rightarrow$ diagonal process

$\Rightarrow \exists \, n_j$ s.t. $\frac{1}{n_j}\sum \varphi_m(f^k(x))$ converges $\forall m$.

For $\varphi \in C(X)$, $\frac{1}{n_j}\sum \varphi(f^k(x))$: converges

$I(\varphi)$

$\varphi(f^{n_j}(x)) - \varphi(x)$

$$\boxed{I(\varphi\circ f) - I(\varphi) = \lim \frac{1}{n_j}\sum_{k=0}^{n_j-1}\left(\varphi(f^{k+1}(x)) - \varphi(f^k(x))\right)}$$

RRT $\Rightarrow \exists$ inv. measure μ.

Q. $\mu = I_x$?

Thm (Birkhoff Ergodic Theorem). $\mu(T^{-1}A) = \mu(A)$.

$T : (X,\mu) \to (X,\mu)$: measure preserving transformation & $\varphi \in L^1(X,\mu)$

For μ-almost every $x \in X$, time average exists ;

$$\varphi_T(x) := \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi(T^k(x))$$

sketch) $\cdot$ $\mathcal{I} := \{A : T^{-1}A = A\}$: invariant σ-algebra.

$\cdot$ $f_{\mathcal{I}} := \dfrac{d(f\mu|_{\mathcal{I}})}{d(\mu|_{\mathcal{I}})}$: (inv) Radon-Nikodym derivative of $f \in L^1(\mu)$.

Conditional expectation.

$\mu(A) = 0 \Rightarrow (f\mu)(A) = 0$

$\int_A f_{\mathcal{I}} \, d\mu|_{\mathcal{I}} = \int_A f \, d\mu|_{\mathcal{I}} \quad \forall A \in \mathcal{I}$.

$$\varphi_{T,n}(x) := \frac{1}{n} \sum_{k=0}^{n-1} \varphi(T^k(x)) \Rightarrow \varphi_T(x) = \lim_{n\to\infty} \varphi_{T,n}(x)$$

$\therefore$ Goal : $\varphi_{T,n}$ converges to $\varphi_{\mathcal{I}}$ a.e. (i.e. $\varphi_T = \varphi_{\mathcal{I}}$ a.e.)

T : ergodic $\Rightarrow$ $\mathcal{I}$-measurable φ is const a.e. $\Rightarrow$ $F_n\varphi \to \int \varphi \, d\mu$.

i.e. $\frac{1}{n} \sum \varphi(T^k(x)) \longrightarrow \int \varphi \, d\mu$ a.e.

- $g := \varphi - \varphi_{\mathcal{I}} - \varepsilon \Rightarrow g_{\mathcal{I}} = \varphi_{\mathcal{I}} - \varphi_{\mathcal{I}} - \varepsilon = -\varepsilon < 0$.
- $F_n := \max_{k \le n} \sum g \circ T^i$. $\quad$ * $\mu(\{x : F_n(x) \nearrow \infty\}) = 0$.
- $F_{n+1} - F_n \circ T = g - \min(0, F_n \circ T)$ $\quad \Downarrow$ $\limsup \frac{F_n}{n} \le 0$ a.e.
- $\limsup_{n\to\infty} \varphi_{T,n}(x) \le \varphi_{\mathcal{I}}(x) + \varepsilon$. $\quad$ ($-\varphi \Rightarrow$ lower bound)
- $\varepsilon \searrow 0 \Rightarrow \varphi_{T,n}(x) \to \varphi_{\mathcal{I}}(x)$ a.e.

Rmk. · conti. f on a metrizable cpt X has an ergodic inv. BP measure.

· μ : invariant BP measure $\Rightarrow$ $\mu = \sum \mu_\alpha$, μ_α ergodic.

$X = \coprod X_\alpha$ (modulo null sets)

· f : uniquely ergodic $\Rightarrow$ $\forall$ conti. φ, $\frac{1}{n}\sum \varphi(f^k(x))$ conv. unif.

$\Leftarrow$ holds if f is topologically transitive.

· U : open, $\mu(\partial U) = 0$ $\Rightarrow$ $\frac{1}{n}\sum_{k=0}^{n-1} \chi_U(f^k(x)) \to \mu(U)$ unif.

Thm (Poincaré Recurrence Theorem).

T : measure preserving transf. of Pro. sp. (X,μ), $A \subset X$: measurable

$\Rightarrow$ $\mu(\{x \in A : \{T^n(x)\}_{n \geq N} \subset X \setminus A\}) = 0$ $\quad \forall N \in \mathbb{N}$.

$N=1$.

sketch) $\tilde{A} := A \cap (\cap T^{-n}(X \setminus A))$

$\Rightarrow$ $\tilde{A} \cap T^{-n}(\tilde{A}) = \phi$ $\Rightarrow$ $T^{-n}(\tilde{A}) \cap T^{-m}(\tilde{A}) = \phi$

$\mu(\tilde{A}) = \mu(T^{-n}\tilde{A})$

$\therefore$ $1 = \mu(X) \geq \mu(\bigcup T^{-n}\tilde{A}) = \sum \mu(T^{-n}\tilde{A}) = \sum \mu(\tilde{A})$ ※

$\{x \in X : \mu(U) > 0 \ \forall \text{open } U \ni x\}$ $\longrightarrow$ · supp μ : closed

· $\mu(X \setminus \text{supp}\,\mu) = 0$

· full measure

$\Rightarrow$ dense in supp μ

Prop. f : conti. on a complete separable metrizable X.

($\not\Leftarrow$) · $\text{supp}\,\mu \subset R(f)$ $\quad \forall$ f-inv. BP measure μ.

· μ : ergodic $\Rightarrow$ $\exists$ dense orbit for $f|_{\text{supp}\,\mu}$. (topological transitivity)

· X : cpt, $f|_{\text{supp}\,\mu}$: uniquely ergodic $\Rightarrow$ $\text{supp}\,\mu$ is a minimal set.

$x \in \alpha(x) \cap \omega(x)$

4.2. Examples of ergodicity ; mixing.

rotation / expanding map / hyperbolic toral automorphism.

* Rotations

Prop. (Kronecker-Weyl Equidistribution Theorem).

Any irrational rotation on circle is uniquely ergodic.

$$\Rightarrow \quad \frac{1}{n}\sum_{k=0}^{n-1}\varphi(R_\alpha^k x) \longrightarrow \int_{S^1}\varphi\, d\theta.$$

sketch) $\{\chi_m(x) = e^{2\pi i m x}\}$

$$m \neq 0, \quad \chi_m(R_\alpha x) = e^{2\pi i m(x+\alpha)} = e^{2\pi i m\alpha}\cdot\chi_m(x)$$

$$\left|\frac{1}{n}\sum_{k=0}^{n-1}\chi_m(R_\alpha^k x)\right| = \frac{1}{n}\left|\sum_{k=0}^{n-1}e^{2\pi i m k\alpha}\right| = \frac{|1-e^{2\pi i m n\alpha}|}{n|1-e^{2\pi i m\alpha}|} \leq \frac{2}{n\cdot c} \to 0$$

- $T_\gamma : \mathbb{T}^n \to \mathbb{T}^n$, $(x_1,\dots,x_n) \mapsto (x_1+\gamma_1,\dots,x_n+\gamma_n)$
- $T_\omega^t : \mathbb{T}^n \to \mathbb{T}^n$: $(x_1,\dots,x_n) \mapsto (x_1+t\omega_1,\dots,x_n+t\omega_n)$.

Prop. $\gamma_1,\dots,\gamma_n,1$: rationally indep. $\Rightarrow$ T_γ : uniquely ergodic (minimal.) w.r.t. Lebesgue measure.

sketch) $$\chi(x_1,\dots,x_n) = \sum_{k_1,\dots,k_n\in\mathbb{Z}} \chi_{k_1,\dots,k_n}\exp(2\pi i\textstyle\sum k_j x_j).$$

$$\Rightarrow \quad \chi_{k_1,\dots,k_n}(1-\exp(2\pi i\textstyle\sum k_j\gamma_j)) = 0$$

$$\Rightarrow \quad \chi_{k_1,\dots,k_n} = 0 \quad \text{except } k_1 = \dots = k_n = 0.$$

* expanding maps

Prop. E_m $(|m|\geq 2)$: ergodic (NOT uniquely.) w.r.t. Lebesgue measure.

sketch) $A \subset S^1$: measurable E_m-inv. set, positive measure. $\exists$ if $\mu(A)<1$.

$\varepsilon>0$, Δ : interval of length $|m|^{-n}$ s.t. $\lambda(\Delta\setminus A) > (1-\varepsilon)\lambda(\Delta) = (1-\varepsilon)|m|^{-n}$

$$\lambda(E_m^n(\Delta)\setminus A) = \lambda(E_m^n(\Delta\setminus A)) = |m|^n\lambda(\Delta\setminus A) > 1-\varepsilon.$$

* mixing.

Def. measure preserving transformation $T : (X, \mu) \to (X, \mu)$: mixing.

if $\mu(T^{-n}(A) \cap B) \to \mu(A) \cdot \mu(B)$ $\forall$ measurable A, B.

Rmk. mixing $\Rightarrow$ ergodic $\quad (\because \mu(T^{-n}(A) \cap (X \setminus A)) = 0)$

Prop. · T_γ : NOT mixing.

· E_m $(|m| \geq 2)$: mixing.

proof) · Δ : suff. small ball in $\mathbb{T}^n$ $\Rightarrow$ $T_\gamma^{-n_k}(\Delta) \cap \Delta = \phi$

· A, B : intervals

$E_m^{-n}(A)$: $|m|^n$ intervals of length $|m|^{-n} \lambda(A)$.

uniformly spread on S^1.

(i.e. $\Delta = \left(\frac{i}{|m|^n}, \frac{i+1}{|m|^n}\right)$ intersects only one cpnt of $E_m^{-n}(A)$)

$n \uparrow$ $\Rightarrow$ B contains approximately $|m|^n \lambda(B)$ cpnts of $E_m^{-n}(A)$.

$\therefore$ $\lambda(E_m^{-n}(A) \cap B) \approx |m|^n \lambda(B) \cdot |m|^{-n} \lambda(A) = \lambda(A) \cdot \lambda(B)$

* hyperbolic toral automorphisms.

Prop. $F_L : \mathbb{T}^2 \to \mathbb{T}^2$: ergodic & mixing w.r.t. Lebesgue measure.

proof) · $\chi_{m,n}(x,y) = \exp 2\pi i(mx+ny)$

$\chi_{m,n}(F_L(x,y)) = \cdots = \chi_{am+cn,\, bm+dn}(x,y)$ $\quad \left(L = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$

$\varphi(x,y) = \sum_{(m,n) \in \mathbb{Z}^2} \varphi_{m,n} \chi_{m,n}(x,y)$

F_L acts on $\mathbb{Z}^2$ by L^t. $\Rightarrow$ $\varphi_{m,n} = \varphi_{am+cn,\, bm+dn}$.

$\to 0$ as $m^2+n^2 \to \infty$

$\therefore$ $\varphi_{m,n} = 0$ $\forall (m,n) \neq (0,0)$.

· Lemma. mixing $\Leftrightarrow$ $\forall \varphi, \psi \in \Phi$: complete system of functions in L^2

$$\int \varphi(T^n x)\, \overline{\psi}(x)\, d\mu \longrightarrow \int \varphi\, d\mu \cdot \int \overline{\psi}\, d\mu \quad \text{as } n \to \infty .$$

$$\int \chi_{m,n}(F_L^N(x,y)) \cdot \overline{\chi_{k,l}(x,y)}\, d\lambda \xrightarrow{N\to\infty} \int \chi_{m,n}\, d\lambda \cdot \int \overline{\chi_{k,l}}\, d\lambda .$$

$m, n, k, l = 0 \quad \Rightarrow \quad 1 \to 1 \cdot 1 .$

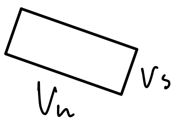
Assume $(m,n) \neq (0,0)$ $\quad \Rightarrow \int \chi_{m,n}\, d\lambda = 0 .$

$\|(L^t)^N (m,n)\| \to \infty \quad \Rightarrow \quad (L^t)^N (m,n) \neq (k,l) \quad$ if $N \gg 0$.

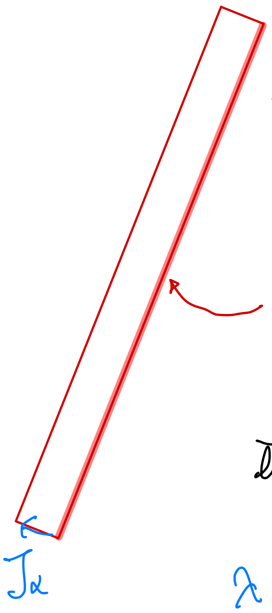
$\Downarrow$

LHS $= 0$.

geometric)



$\leftarrow A$: parallelogram



$\leftarrow F_L^{-n}(A)$: long & thin parallelogram in $\mathbb{R}^2$.

direction (V_s) linear flow : uniquely ergodic.

$J = \{ T_\omega^{-t}(x) \}_{t=0}^{l_n}$

$$\frac{1}{l_n}\int_0^{l_n} \chi_B(T_\omega^{-t}(x))\, dt \to \lambda(B) \quad \text{uniformly} .$$

$$\lambda(F_L^{-n}(A) \cap B) = \int \frac{1}{l_n}\int_0^{l_n} \chi_B(T_\omega^{-t}(x_\alpha))\, dt\, d\alpha \to \lambda(A) \cdot \lambda(B) .$$

*symbolic systems.

· σ_N : mixing w.r.t. μ_p (product measure of probability distribution on $0, \ldots, N-1$)

$$\left(\begin{aligned} & p = (p_0, \ldots, p_{N-1}), \quad 0 \leq p_i \leq 1, \quad \sum p_i = 1 . \\ & \mu_p\left(C_{\alpha_1, \ldots, \alpha_k}^{n_1, \ldots, n_k}\right) = \prod p_{\alpha_i} \end{aligned} \right)$$

4.3. Measure theoretic entropy.

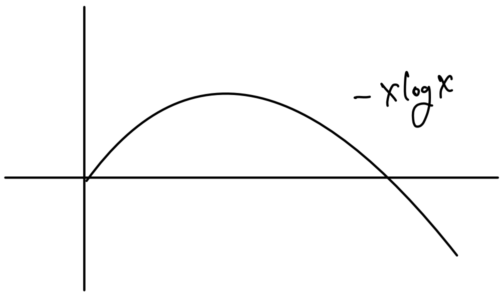
$(X, \mathcal{B}, \mu)$: probability space.

$\xi = \{C_\alpha \in \mathcal{B} : \alpha \in I\}$: measurable partition.

if $\mu(X \setminus \bigcup C_\alpha) = 0$, $\mu(C_{\alpha_i} \cap C_{\alpha_j}) = 0$ $(\alpha_i \neq \alpha_j)$

Def. entropy of ξ ; (Set $0 \cdot \log 0 = 0$)

$$H(\xi) := H_\mu(\xi) := -\sum_{\mu(C_\alpha)>0} \mu(C_\alpha) \log \mu(C_\alpha) \geq 0.$$



containing x & contained in ξ.

$I_\xi : X \to \mathbb{R}$, $I_\xi(x) = -\log \mu(C_\xi(x))$: information function.

$$\Rightarrow H_\mu(\xi) = \int I_\xi \, d\mu.$$

$$0 \leq -\log(\sup \mu(C_\alpha)) \leq H_\mu(\xi) \leq \log|\xi|.$$

Def. conditional entropy of ξ w.r.t. $\eta = \{D_\beta : \beta \in J\}$

$$H(\xi|\eta) := -\sum_\beta \mu(D_\beta) \sum_\alpha \mu(C_\alpha | D_\beta) \log \mu(C_\alpha | D_\beta).$$

$\mu(A|B) = \mu(A \cap B)/\mu(B)$.

$\|$

$\int I_{\xi,\eta} \, d\mu$, $I_{\xi,\eta}(x) = -\log \mu(C_\xi(x) | D_\eta(x))$

conditional information function

$\|$

$\sum_\beta \mu(D_\beta) H(\xi_{D_\beta})$.

ξ_{D_β} : partition of D_β $= \{D_\beta \cap C_\alpha : \alpha \in I\}$

ξ : at most countable w/ finite entropy.

Prop. $\cdot\ 0 \leq H(\xi|\eta) \leq H(\xi)$

refinement. $D\in\eta \Rightarrow \exists C\in\xi$ s.t. $D\subset C$.

$\cdot\ \mu(\xi|\eta) = 0 \iff \xi \leq \eta \pmod 0$

$\cdot\ \mu(\xi|\eta) = H(\xi) \iff \xi,\eta$: indep. $(\mu(C\cap D) = \mu(C)\cdot\mu(D))$

$\cdot\ \zeta \geq \eta \Rightarrow H(\xi|\zeta) \leq H(\xi|\eta)$

$\cdot\ H(\xi\vee\eta|\zeta) = H(\xi|\zeta) + H(\eta|\xi\vee\zeta)$

$\xi\vee\eta = \{C\cap D : C\in\xi,\ D\in\eta,\ \mu(C\cap D)>0\}$.

$H(\xi\vee\eta) = H(\xi) + H(\eta|\xi)$

T: measure preserving transformation, $T^{-1}(\xi) = \{T^{-1}(C_\alpha) : \alpha\in I\}$.

$\xi_{-n}^{T} := \xi \vee T^{-1}(\xi) \vee \cdots \vee T^{-n+1}(\xi)$: joint partition

Def. $h(T,\xi) := h_\mu(T,\xi) = \lim \frac{1}{n} H(\xi_{-n}^{T})$: metric entropy of T rel to ξ.

Prop. $h(T,\xi) = \lim H(\xi | T^{-1}(\xi_{-n}^{T}))$, $H(\xi|T^{-1}(\xi_{-n}^{T}))$: non-increasing.

proof) $H(T^{-1}(\xi_{-n}^{T})\vee\xi) = H(T^{-1}(\xi_{-n}^{T})) + H(\xi|T^{-1}(\xi_{-n}^{T}))$.

$\Rightarrow H(\xi_{-n+1}^{T})/n = H(\xi_{-n}^{T}) + H(\xi|T^{-1}(\xi_{-n}^{T}))$.

$\downarrow$ $h(T,\xi)$

$= \cdots = \cancel{H(\xi_0^T)/n} + \underline{\sum H(\xi|T^{-1}(\xi_{-k}^{T}))/n}$

Def. entropy of T w.r.t. μ;

$h(T) := h_\mu(T) := \sup\{h_\mu(T,\xi) : \xi : \text{measurable partition w/ } H(\xi)<\infty\}$

topological entropy : maximal dynamical complexity

measure theoretic " : average complexity weighted by μ.

Prop. $\cdot\ 0 \leq h(T,\xi) \leq H(\xi)$

$\cdot\ h(T,\xi \vee \eta) \leq h(T,\xi) + h(T,\eta)$

$h(\eta|\xi)=0$

$\Uparrow$

$\cdot\ h(T,\eta) \leq h(T,\xi) + H(\eta|\xi) \quad (\Rightarrow h(T,\eta) \leq h(T,\xi) \text{ if } \eta \leq \xi)$

$\cdot$ (Rokhlin ineq.) $|h(T,\xi) - h(T,\eta)| \leq \underbrace{H(\xi|\eta) + H(\eta|\xi)}_{d_R(\xi,\eta)\,:\ \text{Rokhlin metric}}$.

$\cdot\ h(T,\xi) = h(T, \vee T^{-i}(\xi))$.

Def. $\mathcal{S}$: sufficient family w.r.t. measure preserving T if

← measurable partitions w/ finite entropy.

$\cdot$ (T : non-invertible)

partitions subordinate to $\bigvee_{i=0}^{k} T^{-i}(\xi)$ $\quad (\xi \in \mathcal{S},\ k \in \mathbb{N})$

form a dense subset w.r.t. Rokhlin metric.

$\cdot$ (T : invertible)

same holds for $\bigvee_{i=-l}^{l} T^{i}(\xi)$ $\quad (\xi \in \mathcal{S},\ l \in \mathbb{N})$

Rmk. $\mathcal{S} = \{\xi_n\}_{n\in\mathbb{N}}$

$\sup_{C\in\xi} \operatorname{diam}(C)$.

$\operatorname{diam}(\xi_n) \to 0 \Rightarrow \mathcal{S}$: sufficient.

Def. ξ : generator for T if $\mathcal{S} = \{\xi\}$: suff. family.

Thm. $h_\mu(T) = \sup_{\xi\in\mathcal{S}} h_\mu(T,\xi)$

Cor. ξ : generator for $T \Rightarrow h_\mu(T) = h_\mu(T,\xi)$.

4.4. Examples of calculation of measure theoretic entropy.

* rotations & translations.

$\xi_N = \{[\frac{i}{N}, \frac{i+1}{N}]\} \Rightarrow \{\xi_N\}$: sufficient.

$(\xi_N)_{-n}^{R_\alpha} = \bigvee_{i=0}^{n-1} R_\alpha^{-i}(\xi_N)$ has no more than $N \cdot n$ elts.

$H((\xi_N)_{-n}^{R_\alpha}) \leq \log N \cdot n = \log N + \log n$.

$\Rightarrow h_\lambda(R_\alpha, \xi_N) \leq \lim \frac{1}{n}(\log N + \log n) = 0$.

Similarly, $h_\lambda(T_\gamma, \eta = \xi_N \times \cdots \times \xi_N) = 0$.

* expanding maps.

$\xi_{|k|}$: generator for E_k

$\because \xi_{|k|,n} := \bigvee_{i=0}^{n-1} E_k^{-i}(\xi_{|k|})$: partition w/ equal intervals of length $|k|^{-n}$

entropy $= n \cdot \log |k|$

$\therefore h_\lambda(E_k) = h_\lambda(E_k, \xi_1) = \lim_{n\to\infty} \frac{1}{n} H(\xi_{|k|,n}) = \log |k|$.

* Bernoulli & Markov measures.

μ_p : Bernoulli measure $(p = (p_0, \dots, p_{N-1}),\ \Sigma p_i = 1)$

$\xi_0 = \{C_\alpha^0 : \alpha = 0, \dots, N-1\}$

$\bigvee_{i=-m}^{m} \sigma_N^{i}(\xi_i)$: symm. m cyl. diameter $\to 0$ as $m \to \infty$.

$h_{\mu_p}(\sigma_N) = h_{\mu_p}(\sigma_N, \xi_0) = \lim \frac{1}{n} H\left(\bigvee_{i=0}^{n-1} \sigma_N^{-i}(\xi_0)\right) = H(\xi_0) = -\Sigma p_i \log p_i$.

Cor. $h_{\mu_p}(\sigma_N) \leq h_{top}(\sigma_N) = \log N$ & equality $\Leftrightarrow p = (\frac{1}{N}, \dots, \frac{1}{N})$

$\Pi = (\pi_{ij})$: non-negative matrix

$\uparrow$ stochastic matrix if $\sum_{i=0}^{N-1} \pi_{ij} = 1 \quad j = 0, \dots, N-1$

$\exists$ inv. vector w/ non-negative coords. p $\quad (\Pi p = p)$.

$$\mu_{\pi, p}(C_\alpha^m) = \left(\prod_{i=-m}^{m-1} \pi_{\alpha_i, \alpha_{i+1}}\right) p_{\alpha_m} \quad : \text{Markov measure}$$

Rmk. If Π is transitive, then σ_N is mixing w.r.t. $\mu_{\pi,p}$

$$H(\xi_0 \mid \sigma_N(\xi_0) \vee \cdots \vee \sigma_N^n(\xi_0)) = -\sum_{i,j=0}^{N-1} p_j \pi_{ij} \log \pi_{ij}.$$

$$h_{\mu_{\pi,p}}(\sigma_N) = -\sum_{i,j=0}^{N-1} p_j \pi_{ij} \log \pi_{ij}.$$

$$= \sum_{j=0}^{N-1} p_j \underbrace{H(j\text{-th column})}_{-\sum_{i=0}^{N-1} \pi_{ij} \log \pi_{ij}}$$

A : transitive 0-1 matrix

(Perron-Frobenius) $\Rightarrow$ $\exists\, \lambda_{max}$, pos. eigenvector q.

v : pos. eigenvector of A^T $\longrightarrow$ $\sum q_i v_i = 1$.

$$\pi = \left(\pi_{ij} = \frac{a_{ij} v_i}{\lambda_{max} v_j}\right) \quad : \text{stochastic matrix}$$

Parry measure for TMC σ_A ; $\mu_{\pi,p}$ where $p_i = q_i v_i$.

$$h_{\mu_{\pi,p}}(\sigma_A) = -\sum q_j \cancel{v_j} \frac{a_{ij} v_i}{\lambda_{max} \cancel{v_j}} \log \frac{a_{ij} v_i}{\lambda_{max} v_j}$$

$$= \sum \frac{q_j a_{ij} v_i}{\lambda_{max}} \left(\log \lambda_{max} + \log v_j - \log a_{ij} v_i\right)$$

$$\sum q_j a_{ij} = \lambda_{max} q_j$$

$$= \log \lambda_{max} + \cancel{\sum q_j v_j \log v_j} - \cancel{\sum q_i v_i \log v_i}.$$

$$= \log \lambda_{max}.$$

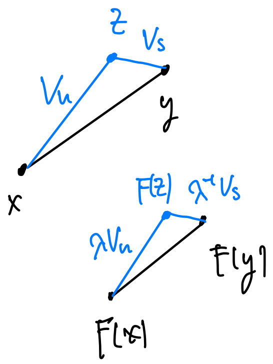
Prop. $h_{\mu_{par}}(\sigma_A) = h_{top}(\sigma_A) = p(\sigma_A) = \log \lambda_{max}$.

$*$ $F = P_L$: hyperbolic toral automorphism.
$\lambda > 1$: unstable

$\Big\} \Rightarrow h_m(F) = \ln \lambda$.

ξ : partition w/ diameter $< \frac{1}{10}$

$C := \bigvee_{k=-n}^{n} F^k(\xi)$



$$d(F^k(x), F^k(z)) = \lambda^{-k} d(x,z) < \frac{\lambda^{-k}}{10}$$

$$d(F^k(x), F^k(z)) \le \cdot + \cdot < \frac{1}{10} + \frac{\lambda^{-k}}{10} < \frac{1}{5}.$$

$$d(x,z) = \lambda^{-n} d(F^n(x), F^n(z)) < \frac{\lambda^{-n}}{5}$$

$$d(y,z) = \lambda^{-n} d(F^{-n}(y), F^{-n}(z)) < \frac{\lambda^{-n}}{5}$$

$\Downarrow$

$$d(x,y) < \frac{2}{5}\lambda^{-n}$$

$\Downarrow$

Measure $\le \frac{2\pi}{5}\lambda^{-2n}$ $\Rightarrow$ lower bound.

$$h(F,\xi) \ge \log \lambda$$

$$h_m(F) \ge \log \lambda = h_{top}(F)$$

$$h_m(F) = \frac{1}{n} F_m(\lambda^n).$$

ξ : suff. partition $\left(\frac{1}{N}\text{-squares}\right)$

$$h_m(F^n, \xi) \leqslant H(\xi \mid F^{-n}(\xi)).$$

$C \in F^{-n}(\xi)$: long & thin parallelogram.

diagram $< \frac{\sqrt{2}}{N} \lambda^n$

width $< \frac{1}{N} \lambda^{-n}$

$\overset{\xi}{D} \cap C \neq \phi \Rightarrow D \subset \frac{\sqrt{2}}{N}$ nbd of C.

area $\leqslant \frac{\sqrt{2}\lambda^n + 2\sqrt{2}}{N} \cdot \frac{2\sqrt{2}+\varepsilon}{N} < \frac{10\lambda^n}{N^2}$

area$(D) = \frac{1}{N^2} \Rightarrow N(C) = \#$ of such D's $< 10\lambda^n$

$$H(\xi \mid F^{-n}(\xi)) < \log \lambda^n + \log 10 = n \log \lambda + \log 10$$

$$\therefore h_m(F) = \frac{1}{n} h_m(F^n) < \log \lambda + \frac{\log 10}{n}$$

$$\Rightarrow h_m(F) \leqslant \log \lambda.$$

4.5. The variational principle.

Thm (Variational principle).

f : homeo on cpt (X, d)

$$h_{top}(f) = \sup_{\mu \in \mathcal{M}(f)} h_\mu(f).$$

$\mathcal{M}(f)$: f-inv BP measures.

$\exists$ positive entropy inv measure

$\Leftrightarrow$ positive topological entropy.

Lemma. $\mu \in \mathcal{M}$

- $x \in X$, $\delta > 0 \Rightarrow \exists \delta' \in (0, \delta)$ s.t. $\mu(\partial B(x, \delta')) = 0$.
- $\delta > 0 \Rightarrow \exists$ finite measurable partition $\xi = \{C_1, \dots, C_k\}$

$$\operatorname{diam} C_i < \delta \quad \& \quad \mu(\partial \xi) = 0.$$

sketch) - $\bigcup_{0<\delta'<\delta} \partial B(x, \delta')$: uncountable union of finite measure

- $B_1, \dots, B_k$: finite cover

$$C_1 := \overline{B_1}, \quad C_i := \overline{B_i} \setminus \overline{B_1} \cup \dots \cup \overline{B_{i-1}}$$

Lemma. $E_n \subset X$: (n, ε)-separated set

$$\nu_n := \frac{1}{\operatorname{card} E_n} \sum_{x \in E_n} \delta_x \quad : \text{uniform } \delta \text{ measure on } E_n$$

$$\mu_n := \frac{1}{n} \sum_{i=0}^{n-1} f_*^i \nu_n$$

$\Rightarrow \exists$ accumulation pt μ of $\{\mu_n\}$ (in the weak* top) s.t. f-inv. &

$$\limsup_{n \to \infty} \frac{1}{n} \log \operatorname{card} E_n \le h_\mu(f)$$

Separated set $\Rightarrow$ entropy $\uparrow$ measure.

sketch) $\xi = \{C_1, \dots, C_k\}$: partition of X.

μ : Borel $\Rightarrow$ $\mu(C_i) = \sup\{\mu(B) : B \subset C_i : \text{closed}\}$.

Choose $B_i \subset C_i$ s.t. $\beta = \{\underset{\text{sea}}{B_0}, \underbrace{\dots, B_k}_{\text{island}}\}$ $(B_0 = X \setminus \cup B_i)$

$H(\xi|\beta) < 1$.

$$h_\mu(f,\xi) \le h_\mu(f,\beta) + H_\mu(\xi|\beta) \le h_\mu(f,\beta) + 1$$

$\mathcal{B} := \{B_0 \cup B_1, \dots, B_0 \cup B_k\}$: open cover of X.

$H_\mu(\beta_n^f) \le \log 2^n \operatorname{card} \beta_n^f$

δ_0 : Lebesgue # of $\mathcal{B}$.

$\downarrow$ " of $\mathcal{B}_n^f$ w.r.t. d_n^f.

$\Downarrow$

δ_0- separated set

$$h_\mu(f,\xi) \le h_\mu(f,\beta) + 1 \le h_{top}(f) + \log 2 + 1.$$

$$h_\mu(f) = \frac{h_\mu(f^n)}{n} \le \frac{h_{top}(f^n) + \log 2 + 1}{n}$$

$\therefore h_\mu(f) \le h_{top}(f)$

Q. $\exists \mu$ s.t. $h_\mu(f) = h_{top}(f)$?

Thm. f : expansive $\Rightarrow$ $\exists$ such a measure μ.

(measure of maximal entropy).